Delta Physics

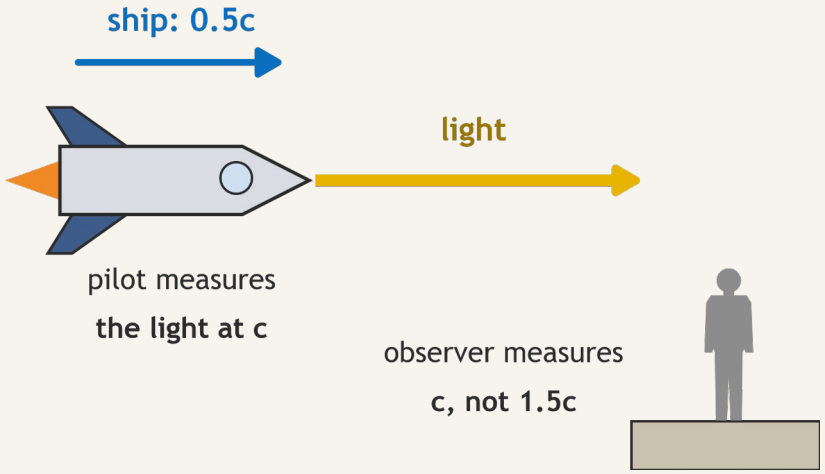
Our Dynamic Universe

Gravitation

5 minutes – Individual

Special Relativity

A ship shines a torch ahead at half the speed of light. How fast is the light?



Higher Physics

PART 1

Frames and the Speed of Light

You walk forward on a moving train. How fast are you going?

By the end of this part…



1. I can **describe** a **frame of reference**
2. I can **determine** a **relative speed** by adding or subtracting
3. I can **state** that the **speed of light** is the **same for all observers**
4. I can **explain** why **time** and **length** must change

Frames of Reference

Frame of reference

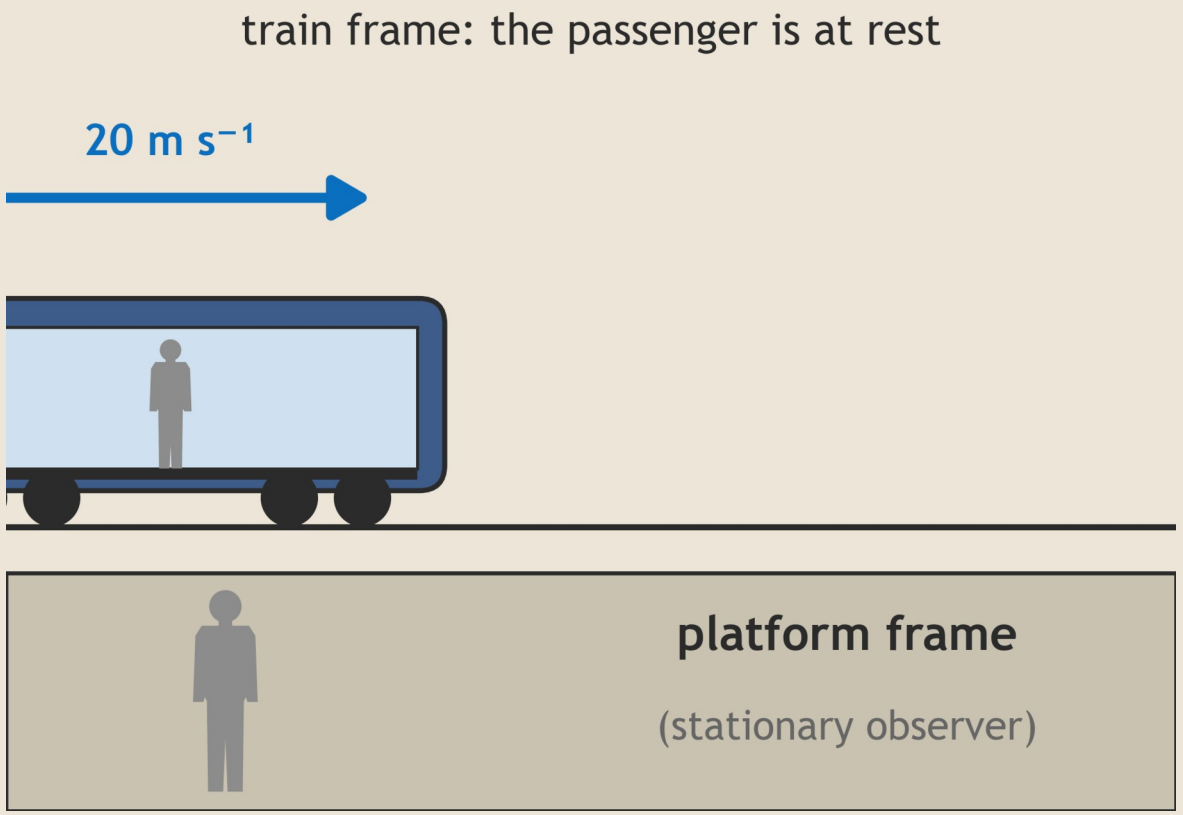
where a speed is **measured from**

Inertial frame

not accelerating: at rest or a **steady speed**

Every speed

is **relative to** a frame



Galilean Relative Speed

To the ground

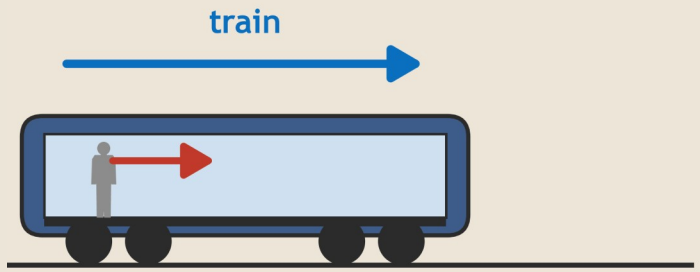
train + walker or **train − walker**

Two trains, same way

subtract the speeds

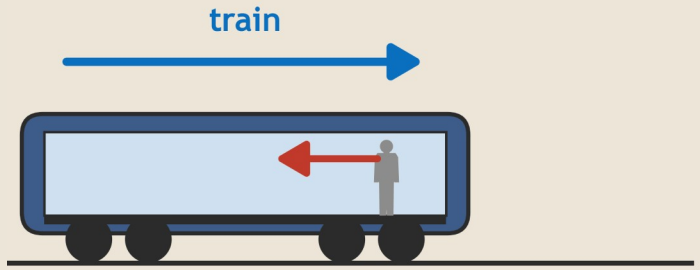
Only for

everyday speeds, far below c



walker

same way: add



walker

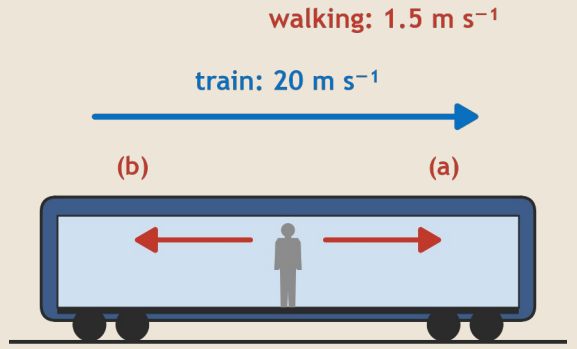
opposite way: subtract

Worked Example

A train travels at 20 m s^{-1} relative to the platform. A passenger walks at 1.5 m s^{-1} relative to the train.

(a) **Determine** the passenger's speed relative to the platform, walking to the front. **1**

(b) **Determine** it when walking to the rear. **1**



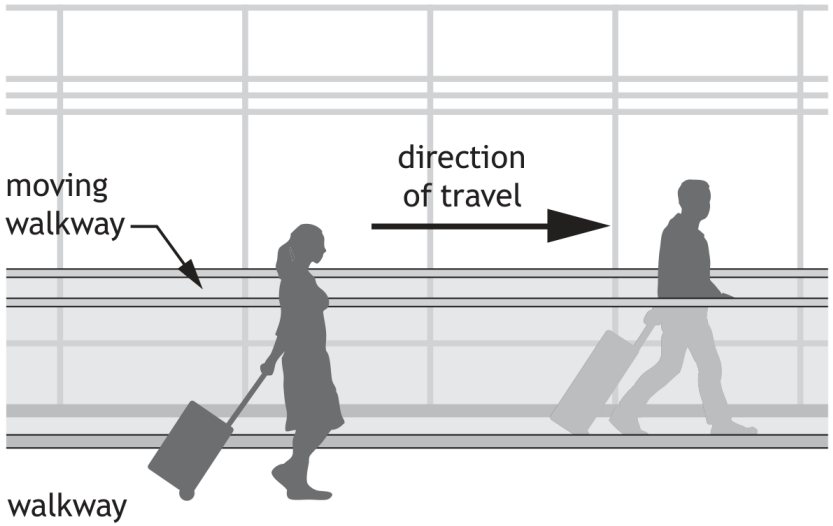
1 mark each: the answer with its unit.

Your Turn: 2016 Q4(a)

4. Two physics students are in an airport building on their way to visit CERN.

(a) The first student steps onto a moving walkway, which is travelling at $0{\cdot}83\,m\,s^{-1}$ relative to the building. This student walks along the walkway at a speed of $1{\cdot}20\,m\,s^{-1}$ relative to the walkway.

The second student walks alongside the walkway at a speed of $1{\cdot}80\,m\,s^{-1}$ relative to the building.



Determine the speed of the first student relative to the second student. **2**

Space for working and answer

Plan: each speed relative to the **building** first

Scan: page 14.

Marking: 2016 Q4(a)

Show Me

1. A car at 30 m s^{-1} overtakes a lorry at 25 m s^{-1}. **Determine** the speed of the car relative to the lorry.

2. Two cars drive towards each other at 30 m s^{-1} and 25 m s^{-1}. **Determine** the speed of one relative to the other.

Show Me

3 A swimmer swims at 1.2 m s^{-1} against a current of 0.5 m s^{-1}. **Determine** the swimmer's speed relative to the bank.

4 A plane flies at 240 m s^{-1} with a tailwind of 20 m s^{-1}. **Determine** its speed relative to the ground.

Past Paper: 2023 Paper 2 Q4(a)(b)

4. Two trains depart from a station at the same time. The trains travel side by side in the same direction, along parallel tracks.

Train A is travelling at 3.5 $m\,s^{-1}$ relative to the platform and train B is travelling at 4.0 $m\,s^{-1}$ relative to the platform.



(a) Determine the speed of train B relative to train A. **1**

Space for working and answer

(b) Once the trains are moving, a passenger on train A walks towards the rear of the train at a speed of 1.3 $m\,s^{-1}$.

Determine the speed of the passenger on train A relative to a passenger seated on train B. **1**

Space for working and answer

Plan: the **rear** of the train is the **opposite way**

Scan: page 12.

Marking: 2023 Paper 2 Q4(a)(b)

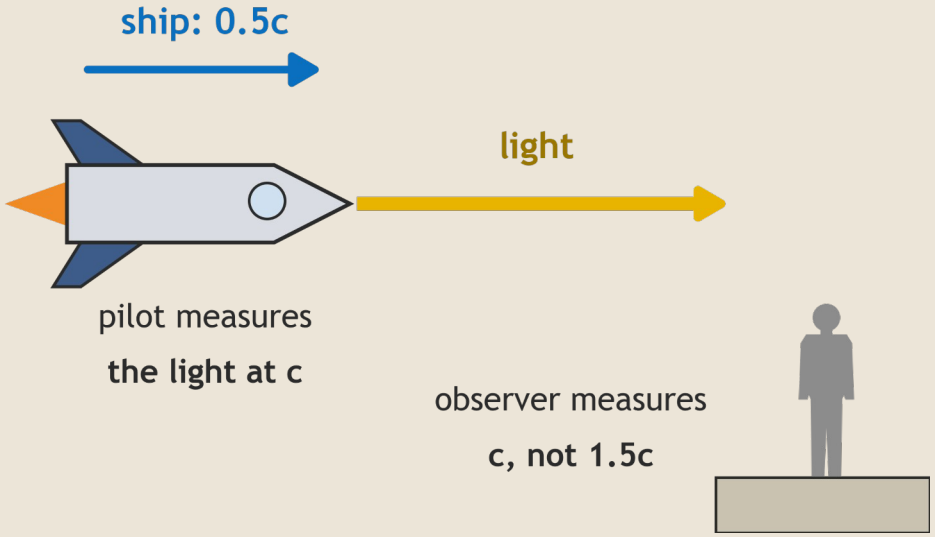
Light Breaks the Rule

Measured

always $c = 3.00 \times 10^{8}\ m\ s^{-1}$

So

the **speed of light** in a vacuum is the **same for all observers**

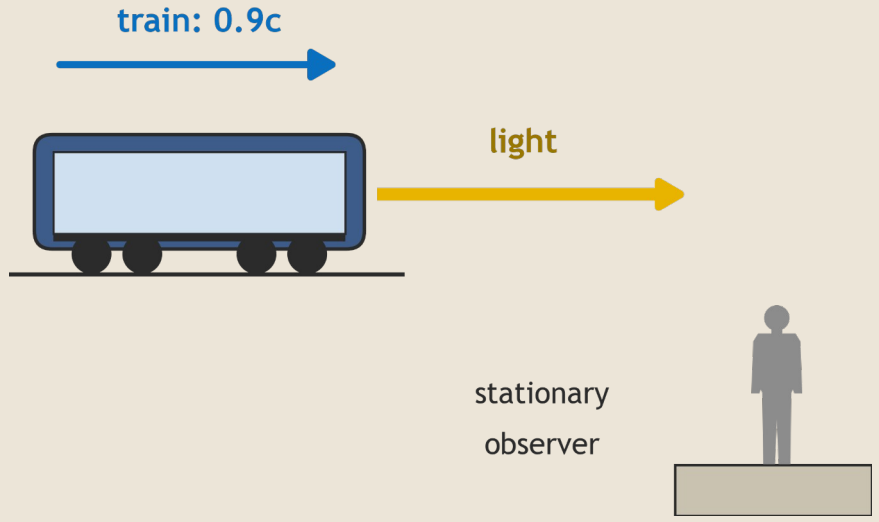


Try it: the speed of light page on the website.

Einstein's Postulates

Postulate 1: the **laws of physics** are the **same** in every **inertial frame of reference**.

Postulate 2: the **speed of light** in a vacuum is the **same for all observers**, however the source or the observer moves.



Time and Length Must Change

Speed

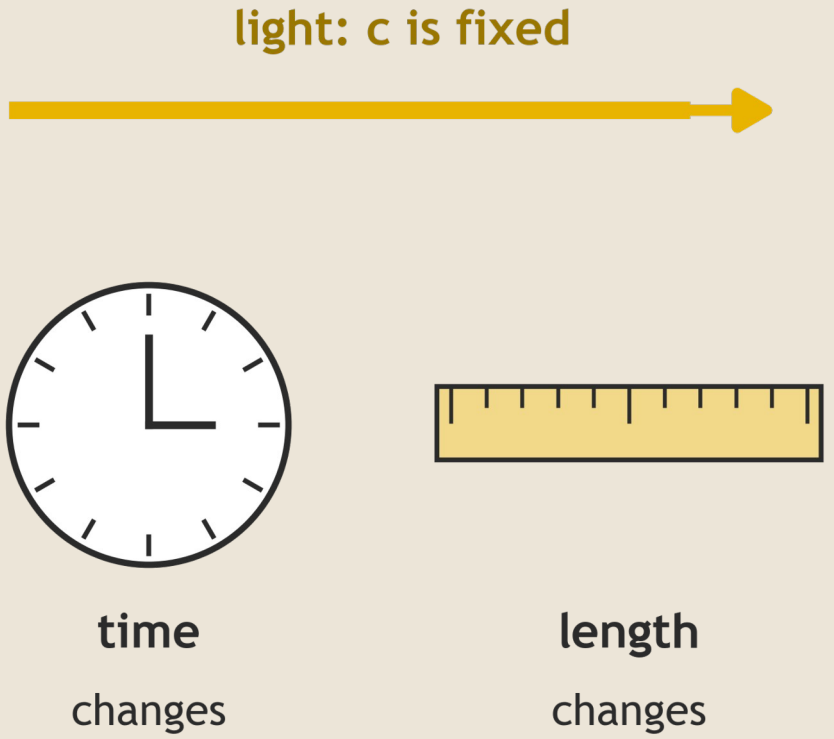
speed = distance ÷ time

c is fixed

so **time** and **distance** must change

Next

time dilation (Part 2) and **length contraction** (Part 3)



Past Paper: 2018 Q6

6. A spacecraft is travelling at $0{\cdot}10c$ relative to a star.

An observer on the spacecraft measures the speed of light emitted by the star to be

A $0{\cdot}90c$

B $0{\cdot}99c$

C $1{\cdot}00c$

D $1{\cdot}01c$

E $1{\cdot}10c$.

Scan: page 5.

Past Paper: 2021 Q9

9. A spacecraft is travelling at $6{\cdot}0 \times 10^7$ m s^{-1} relative to a star.

An observer on the spacecraft measures the speed of light emitted by the star to be

A $2{\cdot}4 \times 10^8$ m s^{-1}

B $2{\cdot}9 \times 10^8$ m s^{-1}

C $3{\cdot}0 \times 10^8$ m s^{-1}

D $3{\cdot}1 \times 10^8$ m s^{-1}

E $3{\cdot}6 \times 10^8$ m s^{-1}.

Scan: page 7.

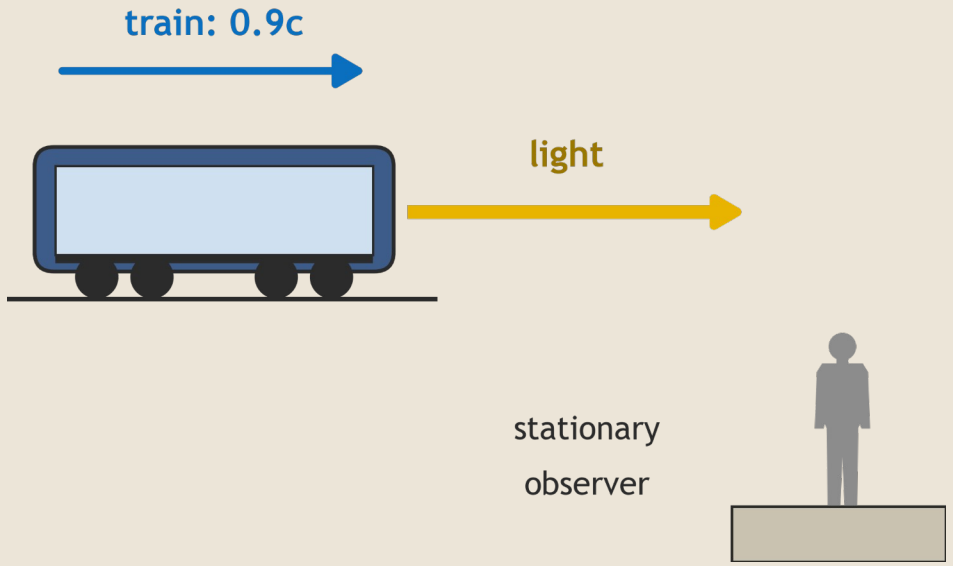
State, Then Justify

State: 3.00×10^{8} m s^{-1} or c

Justify: the speed of light is the same for all observers

A number needs its unit, or 0 marks

Try it: no unit, no answer mark.



Past Paper: 2023 Paper 2 Q4(c)(i)

(c) Two physics students on train A are discussing the possibility of travelling at relativistic speeds. The students consider the train travelling at a speed of $0.9c$ relative to a stationary observer.

(i) The train emits a beam of light towards the stationary observer.

State the speed of the emitted light as measured by the stationary observer.

Justify your answer. **2**

Scan: page 13.

Marking: 2023 Paper 2 Q4(c)(i)

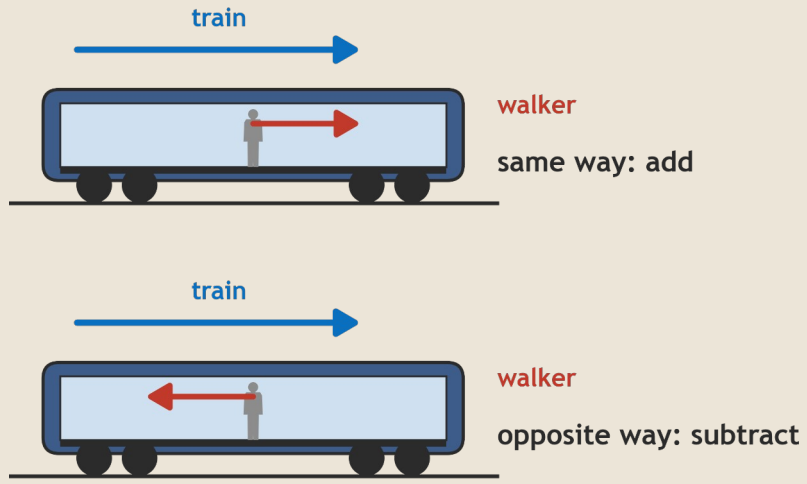
What Candidates Get Wrong (from course reports)

Walking to the rear: subtract; many could not (2023 report Q4(b), 44 %)

A number with no unit: 0 marks (2023 MI Q4(c)(i))

Statement first: if incorrect, 0 marks (2016 MI Q4(b)(i))

Justify the speed of light: 51 % (2023 Q4(c)(i))



Look back — by the end of this part…



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Course specification — 1 of 1

- Knowledge that the speed of light in a vacuum is the same for all observers.
- Knowledge that measurements of space, time and distance for a moving observer are changed relative to those for a stationary observer, giving rise to time dilation and length contraction.

Scan: the course specification, page 8.

Credits (1 of 2)

Your Turn: 2016 Q4(a): © Qualifications Scotland (SQA). 2016 Higher Physics Section 2, Q4

Marking: 2016 Q4(a): © Qualifications Scotland (SQA). 2016 Higher Physics Section 2, Q4

Past Paper: 2023 Paper 2 Q4(a)(b): © Qualifications Scotland (SQA). 2023 Higher Physics Paper 2, Q4; Photograph in 2023 Higher Physics Paper 2, Q4: Iscotlanda Photography/shutterstock.com (the paper's own acknowledgement)

Marking: 2023 Paper 2 Q4(a)(b): © Qualifications Scotland (SQA). 2023 Higher Physics Paper 2, Q4

Past Paper: 2018 Q6: © Qualifications Scotland (SQA). 2018 Higher Physics Section 1, Q6

Past Paper: 2021 Q9: © Qualifications Scotland (SQA). 2021 Higher Physics Paper 1, Q9

Past Paper: 2023 Paper 2 Q4(c)(i): © Qualifications Scotland (SQA). 2023 Higher Physics Paper 2, Q4

Credits (2 of 2)

Marking: 2023 Paper 2 Q4(c)(i): © Qualifications Scotland (SQA). 2023 Higher Physics Paper 2, Q4

Course specification — 1 of 1: © Qualifications Scotland (SQA). Higher Physics Course Specification v4.1 (September 2026), p.8

R Stewart - November 2026